

HW 2.1

AP Calculus AB Unit 2 Notes

2.1 Defining Average and Instantaneous Rates of Change at a Point

2.2 Defining the Derivative of a Function and Using Derivative Notation

Ex 1: Find the slope of...

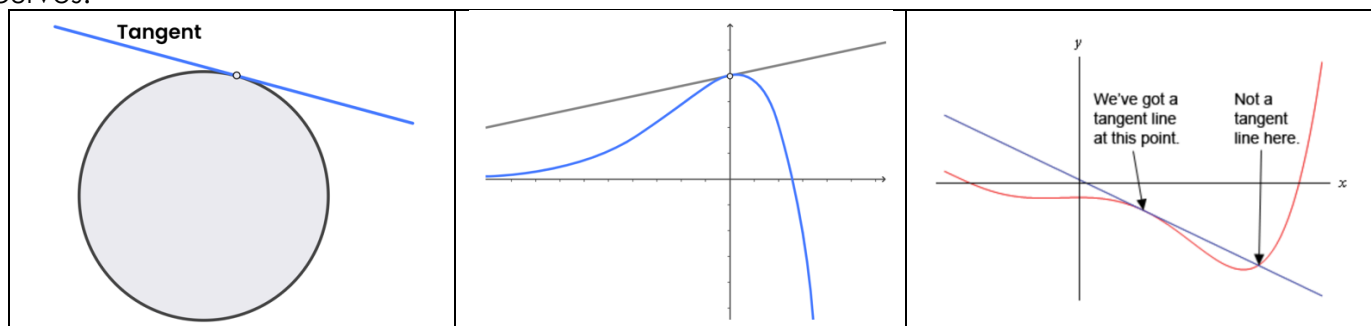
a) $y = 5x + 4$

b) $y - 3 = -2(x + 1)$

c) The line that passes through the points $(0, 3)$ and $(6, -7)$.

What is a tangent line?

A tangent line, or simply tangent, is a line that touches a graph at one point in a small neighborhood around the point of tangency. Tangent lines can be drawn at points on circles or curves.



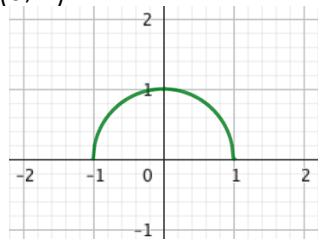
These lines may be tangent to a curve at one point but can later cross the curve again at a point far from the point of tangency.

At the point of tangency the curve and the line share two important properties:

1. _____
2. _____

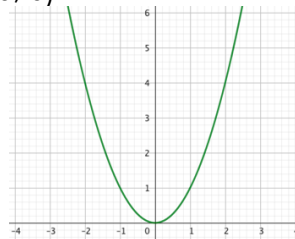
Ex 2: Draw a tangent line at the indicated point. Then, find the slope of the tangent line (and thus the curve) at the indicated point.

a) $(0, 1)$



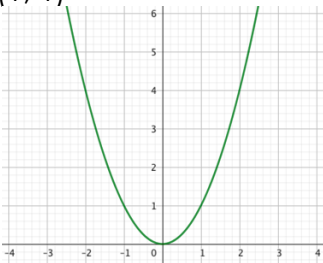
$m =$ _____

b) $(0, 0)$



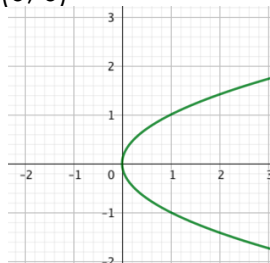
$m =$ _____

c) $(1, 1)$



$m =$ _____

d) $(0, 0)$



$m =$ _____

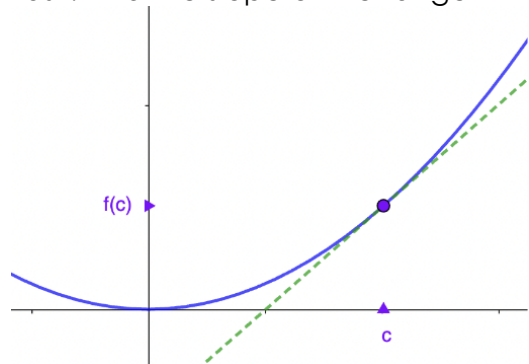
HW 2.1

The Tangent Line Problem

Calculus grew out of four major problems that European mathematicians were working on in the seventeenth century. The first of the four problems is known as the "Tangent Line Problem."



Task: Find the slope of the tangent line to $f(x)$ at $x=c$.



Definition of Tangent Line with Slope m

If f is defined on an open interval containing c , and if the limit

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

exists, then the line passing through $(c, f(c))$ with slope m is the **tangent line** to the graph of f at the point $(c, f(c))$.

Ex 3: Find the slope of f at the given point.

a) $f(x) = x^2 + 1, \quad (2, 5)$	b) $f(x) = x^2 + 1, \quad (-1, 2)$
c) $f(x) = x^2 + 1, \quad (0, 1)$	d) $f(x) = 3, \quad (21, 3)$

Ex 4: Interpret the expression: $\lim_{h \rightarrow 0} \frac{g(6+h) - g(6)}{h} = -5$

HW 2.2

AP Calculus AB Unit 2 Notes

The Derivative of a Function

The **derivative** of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists. For all x for which this limit exists, f' is a function of x .

The notation $f'(x)$ is read _____.

What is a Derivative?

A derivative is _____.

The derivative can also be used to determine the _____ rate of change (or simply the rate of change) of one variable with respect to another.

Vocabulary

- _____ - the process of finding a derivative
- _____ - to find a derivative
- _____ - a derivative exists



"derive" and
"differentiate" are
NOT synonyms!!

Notation

There are three different notations we will use to describe derivatives.

	Type	Notation	Example
1	Lagrange	$f'(x)$	If $f(x) = x^2$, then $f'(x) = 2x$.
2	Leibniz	$\frac{dy}{dx}$	If $y = x^2$, then $\frac{dy}{dx} = 2x$.
3	Leibniz – verb form	$\frac{d}{dx}[f(x)]$	$\frac{d}{dx}[x^2] = 2x$

Ex 5: Find the derivative using the limit process.

a) $f(x) = x^2$	b) $g(x) = \sqrt{x}$	c) $y = \frac{1}{x}$

Ex 6: Find the slope at the given point.

a) $f(x) = x^2$, $x = -2$	b) $g(x) = \sqrt{x}$, $x = 9$	c) $y = \frac{1}{x}$, $x = 10$

Ex 7: Interpret the expression: $h'(20) = 25$

HW 2.2

AP Calculus AB Unit 2 Notes

Alternate Form

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$



Alternate form does NOT yield derivatives, it yields slopes!

Ex 8: Use the alternate form to find the slope at the indicated point.

a) $f(x) = x^2 + 1$, (2, 5)

b) $g(x) = 17 - 4x$, (1, 13)

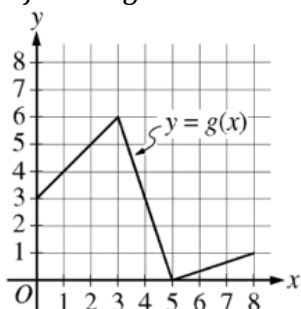
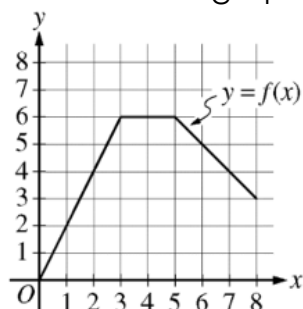
c) $h(x) = |x|$, (0, 0)

Ex 9:

What is $\lim_{h \rightarrow 0} \frac{\cos(\pi + h) - \cos(\pi)}{h}$?

- (A) 1
- (B) $\frac{\sqrt{2}}{2}$
- (C) 0
- (D) -1
- (E) The limit does not exist.

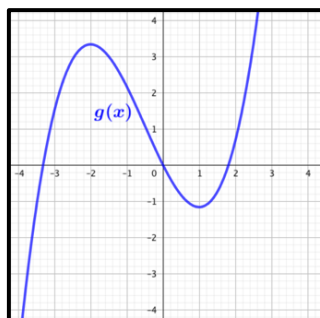
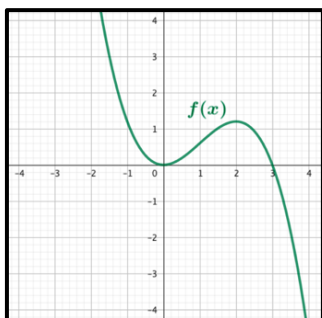
Ex 10: Use the graphs of f and g below to evaluate each expression.



a) $\lim_{x \rightarrow 4} \frac{g(x) - g(4)}{x - 4}$

b) $\lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$

Ex 11: Use the graphs of f and g to arrange the values from least to greatest:



I. $g'(-1)$

II. $f'(1)$

III. $f'(0)$

HW 2.3

AP Calculus AB Unit 2 Notes

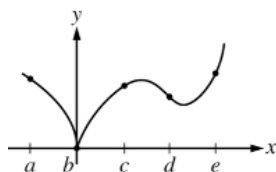
2.3 Estimating Derivatives of a Function at a Point

2.4 Connecting Differentiability and Continuity: Determining When Derivatives Do and Do Not Exist

Ex 1: What is the average rate of change of $y = \cos(2x)$ on the interval $\left[0, \frac{\pi}{2}\right]$?

- (A) $-\frac{4}{\pi}$ (B) -1 (C) 0 (D) $\frac{\sqrt{2}}{2}$ (E) $\frac{4}{\pi}$

Ex 2: The graph of the function f is shown in the figure. For which of the following values of x is $f'(x)$ positive and increasing?



Graph of f

- (A) a
(B) b
(C) c
(D) d
(E) e

Ex 3: Use the alternate form to find the slope at the indicated point, if possible.

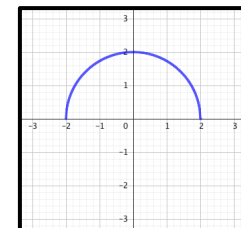
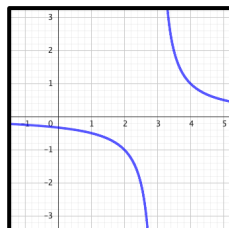
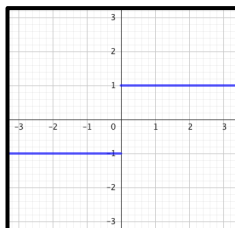
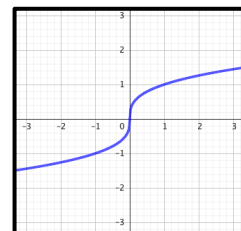
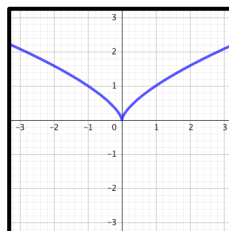
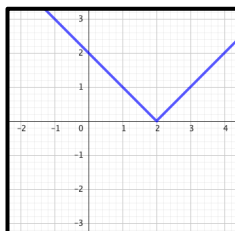
a) $g(x) = \sqrt[3]{x}$, $(0, 0)$

b) $h(x) = \sqrt{x}$, $(0, 0)$

Differentiability and Continuity

Functions are *not* differentiable at

1. _____
2. _____
3. _____
4. _____



A consequence of differentiability is _____. That is, if f is differentiable at $x = c$, f is also _____ at $x = c$.

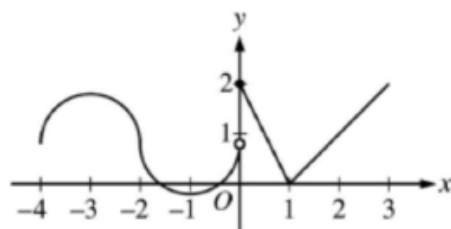


A continuous function may NOT be differentiable!

HW 2.3

AP Calculus AB Unit 2 Notes

Ex 4:



Graph of f

- (A) $x = 1$
- (B) $x = -2$ and $x = 0$
- (C) $x = -2$ and $x = 1$
- (D) $x = 0$ and $x = 1$

The graph of the piecewise-defined function f is shown in the figure above. The graph has a vertical tangent line at $x = -2$ and horizontal tangent lines at $x = -3$ and $x = -1$. What are all values of x , $-4 < x < 3$, at which f is continuous but not differentiable?

Ex 5: If g is a function for which $\lim_{h \rightarrow 0} \frac{g(3+h) - g(3)}{h} = 0$, which of the following statements must be true?

- | | | |
|----------------|----------------------------------|------------------------------------|
| I. $g'(3) = 0$ | II. g is continuous at $x = 3$ | III. g' is continuous at $x = 3$ |
|----------------|----------------------------------|------------------------------------|

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only
- (E) I, II and III

Approximating Derivatives with Tables

To approximate a derivative of f at $x = c$ using tabular data, find the smallest interval containing $x = c$, then find the _____ on that interval.

Ex 6: The table shows $f(x)$ for several values of x , where f is a differentiable function of x . Estimate the instantaneous rate of change of f at $x = 5$.

x	0	2	4	6
$f(x)$	0	10	35	75

Ex 7: **2024 AB 1 part (a)**

t (minutes)	0	3	7	12
$C(t)$ (degrees Celsius)	100	85	69	55

The temperature of coffee in a cup at time t minutes is modeled by a decreasing differentiable function C , where $C(t)$ is measured in degrees Celsius. For $0 \leq t \leq 12$, selected values of $C(t)$ are given in the table shown.

- (a) Approximate $C'(5)$ using the average rate of change of C over the interval $3 \leq t \leq 7$. Show the work that leads to your answer and include units of measure.

HW 2.3

AP Calculus AB Unit 2 Notes

Approximating Derivatives Using Technology

Ex 8: Use a graphing utility to find the indicated derivative.

a) $f(x) = \ln(x^2 - 4x + 5)$, $f'(1)$

b) $y = 7.6 \tan^{-1}(0.2x)$, $\left. \frac{dy}{dx} \right|_{x=6}$

Ex 9: The rate of flow of gasoline, in gallons per second, can also be modeled by

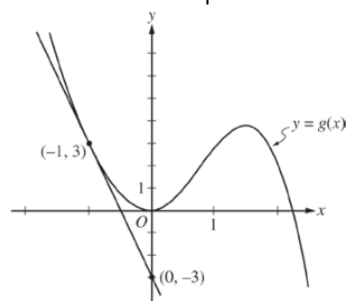
$g(t) = \left(\frac{t}{450}\right) \cos\left(\left(\frac{t}{100}\right)^2\right)$ for $0 \leq t \leq 150$. Using this model, find the value of $g'(140)$. Indicate units of measure.

Writing Equations of Tangent Lines

To write an equation of a tangent line, you need a _____ and a _____.

Ex 10: Write an equation of the tangent line to $f(x) = \ln(x^2 - 4x + 5)$ at $x = 1$.

Ex 11: The figure shows the graph of the function g and the line tangent to the graph of g at $x = -1$. Write an equation of the tangent line to g at $x = -1$.



HW 2.4

AP Calculus AB Unit 2 Notes

2.5 Applying the Power Rule

2.6 Derivative Rules: Constant, Sum, Difference, and Constant Multiple

Ex 1: On your TI84 calculator, press **app** and scroll down (or up) until you find the **Transform** app and press enter. Press **y=** and type in:



$$Y_1 = x^A \quad Y_2 = \frac{d}{dx}(Y_1)|_{x=x}$$

Press **graph**. Using the left and right arrows change the value of A to graph the derivatives of...

a) $y = x^2$ ($A = 2$)	b) $y = x^3$ ($A = 3$)	c) $y = x^4$ ($A = 4$)	d) $y = x^5$ ($A = 5$)
--------------------------	--------------------------	--------------------------	--------------------------

What relationship do you notice about the degrees of y and y' ?

Shortcut Rules

In the last lesson, you used the limit definition to find derivatives. In this and the next few, you will be introduced to several "shortcut rules" that allow you to find derivatives *without* the use of the limit definition.

1. Constant Rule: $\frac{d}{dx}[c] = \underline{\hspace{2cm}}$

Ex 2: Differentiate.

a) $y = 12345\pi$	b) $f(x) = \sqrt{9}$
-------------------	----------------------

2. Power Rule: $\frac{d}{dx}[x^n] = \underline{\hspace{2cm}}$

Ex 3: Differentiate.

a) $f(x) = x^3$	b) $y = x^{20}$	c) $g(x) = \frac{1}{x}$
d) $h(x) = \sqrt{x}$	e) $f(x) = \frac{1}{\sqrt{x}}$	

HW 2.4

AP Calculus AB Unit 2 Notes

3. Constant Multiple Rule: $\frac{d}{dx}[cf(x)] = \underline{\hspace{2cm}}$

Ex 4: Differentiate.

a) $s(t) = 20t^7$	b) $y = \frac{4}{\sqrt[3]{x}}$
c) $y = \pi$	d) $y = 10x$

4. Sum and Difference Rules: $\frac{d}{dx}[f(x) \pm g(x)] = \underline{\hspace{2cm}}$

Ex 5: Differentiate.

a) $f(x) = 3x^4 - 2x + \pi$	b) $y = \pi^2 + \frac{1}{\pi} - \sqrt{\pi}$	c) $g(x) = \frac{x}{5} - \frac{5}{x}$
d) $v(t) = (t + 1)(3t - 7)$	e) $y = (x - 5)^2$	
f) $y = \frac{3x^4 + 2x^2 - 5}{x^2}$	g) For the gifted... Let $\star \in \mathbb{Q}$ $y = 5x^{\star} - \frac{2}{x^{\star-1}}$	

HW 2.4

AP Calculus AB Unit 2 Notes

Ex 6: Write an equation of the tangent line to $f(x) = \frac{2}{\sqrt[5]{x}}$ at the point $(1, 2)$.

Ex 7: Determine the points, if any, at which $g(x) = 2x^3 + 3x^2 - 12x + 5$ has a horizontal tangent line.

Ex 8:

Let f be the function given by $f(x) = x^3 - 6x^2 + 8x - 2$. What is the instantaneous rate of change of f at $x = 3$?

- (A) -5 (B) $-\frac{15}{4}$ (C) -1 (D) 6 (E) 17

Ex 9:

$$\lim_{x \rightarrow e} \frac{(x^{20} - 3x) - (e^{20} - 3e)}{x - e} \text{ is}$$

- (A) 0 (B) $20e^{19} - 3$ (C) $e^{20} - 3e$ (D) nonexistent

Ex 10: Let f be the piecewise-linear function defined. Which of the following statements are true?

$$f(x) = \begin{cases} 2x - 2 & \text{for } x < 3 \\ 2x - 4 & \text{for } x \geq 3 \end{cases}$$

I. $\lim_{h \rightarrow 0^-} \frac{f(3+h) - f(3)}{h} = 2$

II. $\lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} = 2$

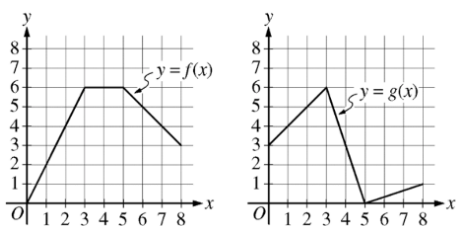
III. $f'(3) = 2$

- (A) None
(B) II only
(C) I and II only
(D) I, II and III

HW 2.4

AP Calculus AB Unit 2 Notes

Ex 11: The graphs of the piecewise linear functions f and g are shown.



a) The function h is defined by $h(x) = f(x) + 3g(x)$, find $h'(2)$ or state why it doesn't exist.

b) The function p is defined by $p(x) = 6x^2 - 5f(x)$, find $p'(3)$ or state why it doesn't exist.

Ex 12: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x .

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

a) Write an equation of the tangent line to $m(x) = 2f(x) + 3$, at $x = 7$.

b) The function n is defined by $n(x) = f(x) + 2g(x)$. Determine the x -values, if any, at which $n(x)$ has a horizontal tangent line.

Ex 13: Find an equation of the line that is tangent to $f(x) = 5x^2 - 3$ and parallel to $5x - y = 4$.

Ex 14: Find k such that the line $y = -2x + 3$ is tangent to $f(x) = kx^2$.

HW 2.5

AP Calculus AB Unit 2 Notes

2.7 Derivatives of $\sin x$, $\cos x$, e^x and $\ln x$

2.8 The Product Rule

Ex 1:

Let f be the function given by $f(x) = x^3 + 5x$. For what value of x in the closed interval $[1, 3]$ does the instantaneous rate of change of f equal the average rate of change of f on that interval?

- (A) $\sqrt{\frac{7}{3}}$ (B) $\sqrt{\frac{13}{3}}$ (C) $\sqrt{5}$ (D) $\sqrt{6}$ (E) $\sqrt{\frac{19}{3}}$

Ex 2: Fill in the blank.

- a) A function, $f(x)$, is continuous at $x = a$ if _____.
- b) A function is differentiable at $x = a$ if $f(x)$ is _____ at $x = a$ and _____.

Ex 3:

At $x = 3$, the function given by $f(x) = \begin{cases} x^2, & x < 3 \\ 6x - 9, & x \geq 3 \end{cases}$ is

- (A) undefined.
(B) continuous but not differentiable.
(C) differentiable but not continuous.
(D) neither continuous nor differentiable.
(E) both continuous and differentiable.

Ex 4: Differentiate.

a) $f(x) = \begin{cases} x^2, & x \leq 1 \\ 1 + 2x, & x > 1 \end{cases}$

b) $f(x) = |x + 4|$

Ex 5: Find the slope of $f(x) = |x + 4|$ at the indicated x -value, or explain why it does not exist.

a) $x = 0$

b) $x = -5$

c) $x = -4$

HW 2.5

AP Calculus AB Unit 2 Notes

Ex 6: Let f be the piecewise-linear function defined. Which of the following statements are true?

$$f(x) = \begin{cases} 2x - 2 & \text{for } x < 3 \\ 2x - 4 & \text{for } x \geq 3 \end{cases}$$

I. $\lim_{h \rightarrow 0^-} \frac{f(3+h) - f(3)}{h} = 2$

II. $\lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} = 2$

III. $f'(3) = 2$

(A) None

(B) II only

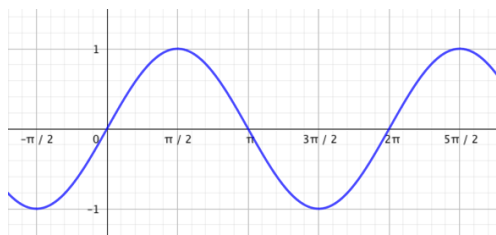
(C) I and II only

(D) I, II and III

Ex 7: Find a and b so that $f(x) = \begin{cases} 2x - x^2, & x \leq 1 \\ x^2 + ax + b, & x > 1 \end{cases}$ is differentiable.

Ex 8: Below are the graph of $f(x) = \sin x$.

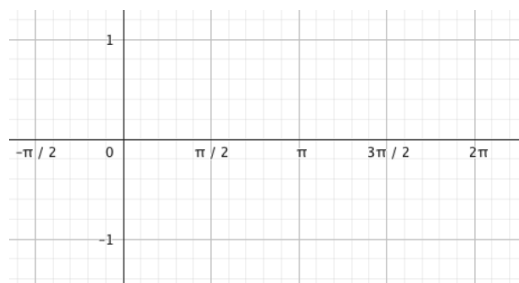
a) Use the graph to fill in the table.



x	$f'(x)$
0	
$\frac{\pi}{2}$	
π	
$\frac{3\pi}{2}$	
2π	

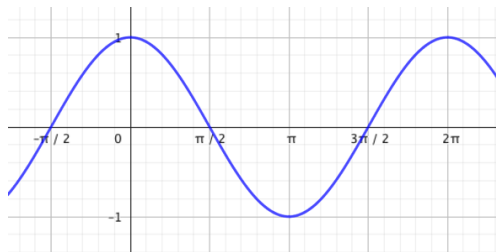
b) Plot the points $(x, f'(x))$ and connect the dots. This is the graph of the derivative of $f(x) = \sin x$. Look familiar?

$$f'(x) = \underline{\hspace{2cm}}$$



Ex 9: Below are the graph of $g(x) = \cos x$.

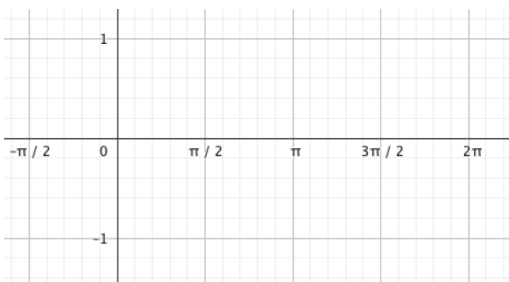
a) Use the graph to fill in the table.



x	$g'(x)$
0	
$\frac{\pi}{2}$	
π	
$\frac{3\pi}{2}$	
2π	

b) Plot the points $(x, g'(x))$ and connect the dots. This is the graph of the derivative of $g(x) = \cos x$. Look familiar?

$$g'(x) = \underline{\hspace{2cm}}$$



HW 2.6

AP Calculus AB Unit 2 Notes

Differentiation Rules – Sine and Cosine

$$\frac{d}{dx} [\sin x] = \underline{\hspace{2cm}} \quad \frac{d}{dx} [\cos x] = \underline{\hspace{2cm}}$$

Ex 10: Differentiate: $f(x) = 3 \cos x + 2$

Ex 11: What is the slope of the tangent line to the graph of $f(x) = 3 \cos x + 2$ at the point $x = \frac{3\pi}{2}$?

Ex 12: Consider the function $h(x) = e^x$.

a) Use a graphing utility to fill in the table.

x	$h(x)$	$h'(x)$
-2		
-1		
0		
1		
2		

b) Compare the $h(x)$ and $h'(x)$ values in the table. What do you notice?

c) $h'(x) = \underline{\hspace{2cm}}$



Differentiation Rules – Exponential Functions

$$\frac{d}{dx} [e^x] = \underline{\hspace{2cm}} \quad \frac{d}{dx} [a^x] = \underline{\hspace{2cm}}$$

Ex 13: Differentiate: $f(x) = 5e^x + \sqrt{x}$

Differentiation Rules – Logarithmic Functions

$$\frac{d}{dx} [\ln x] = \underline{\hspace{2cm}} \quad \frac{d}{dx} [\log_a x] = \underline{\hspace{2cm}}$$

Ex 14: Differentiate: $g(x) = 7 \sin x + \ln x - 2 \cdot 5^x + \frac{8}{x^3}$

Product Rule

$$\frac{d}{dx} (f(x) \cdot g(x)) = \underline{\hspace{2cm}}$$

An easy way to remember the product rule: derivative sandwich!



HW 2.7

AP Calculus AB Unit 2 Notes

Ex 15: Find the derivative.

a) $y = (5x + 1) \ln x$

b) $f(x) = x^5 \cos x$



Not all products must be differentiated with the product rule. Sometimes it is easier or quicker to REWRITE first!

Ex 16: Which equations **must** be differentiated with the product rule?

a) $f(x) = 4 \sin x$

b) $y = xe^x$

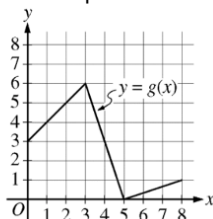
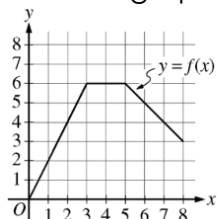
c) $y = (x + 1)(x^2 - 3)$

d) $g(x) = 2x^3$

e) $y = 2 \sin x \cos x$

Ex 17: The graphs of the piecewise linear functions f and g are shown.

If $h(x) = f(x) \cdot g(x)$, find $h'(7)$ or state that it does not exist.



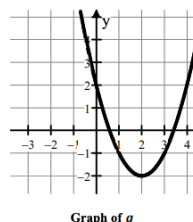
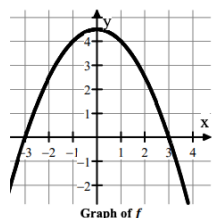
Ex 18: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x .

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

a) If $h(x) = f(x) \cdot g(x)$, find $h'(0)$ or state that it does not exist.

b) Write a tangent line to $h(x)$ at $x = 0$.

Ex 19: The graphs of two differentiable functions f and g are shown below. If $h(x) = f(x) \cdot g(x)$, which of the following statements about $h'(3)$ are true?



a) $h'(3) < 0$

b) $h'(3) = 0$

c) $h'(3) > 0$

d) $h'(3)$ is undefined

e) There is not enough information to conclude anything about $h'(3)$.

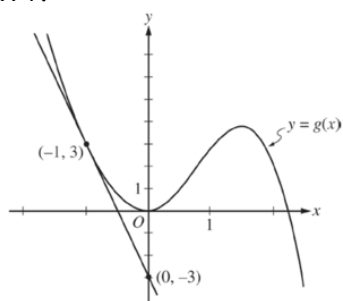
HW 2.8

AP Calculus AB Unit 2 Notes

2.9 The Quotient Rule

2.10 Finding the Derivatives of Tangent, Cotangent, Secant, and Cosecant Functions

Ex 1:



The figure above shows the graph of the function g and the line tangent to the graph of g at $x = -1$. Let h be the function given by $h(x) = e^x \cdot g(x)$. What is the value of $h'(-1)$?

- (A) $\frac{9}{e}$ (B) $\frac{-3}{e}$ (C) $\frac{-6}{e}$ (D) $\frac{-6}{e} - \frac{3}{e^2}$ (E) -6

Quotient Rule

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \underline{\hspace{2cm}}$$



Ex 2: If $h(x) = \frac{x}{3 \cos x + 1}$, find $h'(\pi)$.

Ex 3: Write a tangent line to $h(x) = \frac{x}{3 \cos x + 1}$ at $x = \pi$.



Not all quotients must be differentiated with the quotient rule. Sometimes it is easier or quicker to REWRITE first!

Ex 4: Which equations **must** be differentiated with the quotient rule?

- a) $f(x) = \frac{5}{x}$
- b) $y = \frac{x}{5}$
- c) $y = \frac{e^x}{x+1}$
- d) $g(x) = \frac{x+1}{x}$
- e) $y = \frac{\sin x}{\cos x}$

HW 2.9

AP Calculus AB Unit 2 Notes

Ex 5: Find the derivative.

a) $y = \tan x$

b) $y = \cot x$

c) $y = \sec x$

d) $y = \csc x$

Trigonometric Derivatives

$$\frac{d}{dx} [\sin x] = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} [\cos x] = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} [\tan x] = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} [\csc x] = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} [\sec x] = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} [\cot x] = \underline{\hspace{2cm}}$$

A Helpful Chart

$$\tan x$$

$$\sec x$$

$$\sec x$$

$$\cot x$$

$$\csc x$$

$$-\csc x$$



*Cover the term you wish to differentiate. The derivative is the product of the remaining terms in the same row.

For example, the derivative of tangent is $\sec x \cdot \sec x = \sec^2 x$

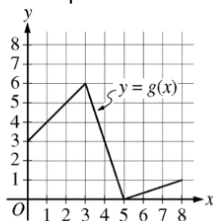
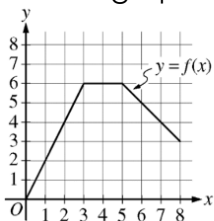


$\sec x$

$\sec x$

Ex 6: The graphs of the piecewise linear functions f and g are shown.

If $m(x) = \frac{f(x)}{g(x)}$, find $m'(6)$ or state that it does not exist.



Ex 7: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x .

If $m(x) = \frac{g(x)}{f(x)}$, find $m'(7)$ or state that it does not exist.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

HW 2.9

AP Calculus AB Unit 2 Notes

Ex 8: Find the point(s) at which $f(x) = \frac{x^2}{x^2+1}$ has a horizontal tangent.

Ex 9:

The function f is defined by $f(x) = \frac{x}{x+2}$. What points (x,y) on the graph of f have the property that the line tangent to f at (x,y) has slope $\frac{1}{2}$?

- (A) $(0,0)$ only
- (B) $\left(\frac{1}{2}, \frac{1}{5}\right)$ only
- (C) $(0,0)$ and $(-4,2)$
- (D) $(0,0)$ and $\left(4, \frac{2}{3}\right)$
- (E) There are no such points.

Ex 10:

The graph of $y = e^{\tan x} - 2$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

- (A) 0.606
- (B) 2
- (C) 2.242
- (D) 2.961
- (E) 3.747

